NON-BIPARTITE PAIRS OF 3-CONNECTED GRAPHS ARE HIGHLY RAMSEY-INFINITE

MARK SIGGERS

ABSTRACT. A pair of graphs $(H_\mathrm{b}, H_\mathrm{r})$ is highly Ramsey-infinite if there is some constant c such that for large enough n there are at least 2^{cn^2} non-isomorphic graphs on n or fewer vertices that are minimal with respect to the property that when their edges are coloured blue or red, there is necessarily a blue copy of H_b or a red copy of H_r.

We show that a pair of 3-connected graphs is highly Ramsey-infinite if and only if at least one of the graphs in non-bipartite. Further we show that the pair $(H_\mathrm{b}, H_\mathrm{r})$ is highly Ramsey infinite for H_r an odd cycle of girth ℓ and H_b any graph with no induced cycle of length ℓ or longer.

In showing the above results, we continue the theory of gadgets called senders and determiners that has been developed over many earlier papers on Ramsey-infinite graphs.

1. INTRODUCTION

All graphs in this paper are simple, finite and undirected. For graphs G, H_b, and H_r, we write $G \to (H_\mathrm{b}, H_\mathrm{r})$ and say G is *Ramsey* for $(H_\mathrm{b}, H_\mathrm{r})$, to mean that any blue-red colouring of the edges of G results in a blue copy of H_b or a red copy of H_r. By a *copy* of a graph H in a graph G we mean a not necessarily induced subgraph of G isomorphic to the graph H. The pair $(H_\mathrm{b}, H_\mathrm{r})$ is called *asymmetric* if H_b and H_r are non-isomorphic.

A graph is *Ramsey-minimal* for $(H_\mathrm{b}, H_\mathrm{r})$ if it is Ramsey for $(H_\mathrm{b}, H_\mathrm{r})$ but no proper subgraph is. A pair $(H_\mathrm{b}, H_\mathrm{r})$ is *Ramsey-infinite* if there infinitely many graphs that are Ramsey-minimal for $(H_\mathrm{b}, H_\mathrm{r})$. It is *highly Ramsey-infinite* if for some constant c, and large enough n, there are at least 2^{cn^2} graphs on at most n vertices that are Ramsey-minimal for $(H_\mathrm{b}, H_\mathrm{r})$. We use the word 'highly', as this is best possible up to the constant.

1.1. Ramsey-infinite pairs.

In [14], Nešetřil and Rödl started characterising the pairs $(H_\mathrm{b}, H_\mathrm{r})$ that are Ramsey-infinite. In [3], Burr et. al showed that $(H_\mathrm{b}, H_\mathrm{r})$ is Ramsey-finite if H_b is a matching. In [4], the same authors showed that it is Ramsey-finite in the case that H_b and H_r are both odd stars. These seem to be the only Ramsey-finite pairs.

Over several papers, Burr et. al [5, 6, 7], Faudree [11], and Łuczak [13], settled this except for when H_b and H_r are both non-forests. That is, if at least one of H_b and H_r is a forest, and is not one of the finite cases mentioned above, then (H_b, Hr) is Ramsey-infinite. Rödl and Ruciński settled the remaining symmetric cases in [15].

The problem is open in the case that H_b and H_r are non-isomorphic non-forests, but large sub-cases are known. In [14] it was shown that $(H_\mathrm{b}, H_\mathrm{r})$ is Ramsey-infinite if H_b and H_r both contain an odd cycle or are 3-connected. The most recent significant result is a

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result of Bollobás et.al. [1], where they show that for $H_{\rm r}$ a cycle of length ℓ, and $H_{\rm b}$ a graph containing no induced cycle of length ℓ or more, $(H_{\rm b}, H_{\rm r})$ is Ramsey-infinite.

1.2. **Highly Ramsey-infinite pairs.** In [10], the authors considered a stronger notion of Ramsey-infinite, and showed that when $H_{\rm b}$ and $H_{\rm r}$ are 3-connected there are at least $2^{cn\log n}$ different graphs on n or fewer vertices that are Ramsey-minimal for $(H_{\rm b}, H_{\rm r})$.

In [16] we improved this in the case that $H_{\rm b}$ and $H_{\rm r}$ are cliques to show that $(H_{\rm b}, H_{\rm r})$ is highly Ramsey-infinite. In [17, 18], this was extended to $H_{\rm b} = H_{\rm r}$ being an odd cycle. Further, in [17], we made the observation that if $H_{\rm b}$ and $H_{\rm r}$ are bipartite, then $(H_{\rm b}, H_{\rm r})$ cannot be highly Ramsey-infinite.

In this paper, we find many more pairs of highly Ramsey-infinite graphs. In particular, we show (Corollary 7.2) that for 3-connected $H_{\rm b}$ and $H_{\rm r}$, $(H_{\rm b}, H_{\rm r})$ is highly Ramsey-infinite if and only if at least one of $H_{\rm b}$ and $H_{\rm r}$ is non-bipartite. Using the proof from [1], we show (Corollary 5.2) that if $H_{\rm b}$ is a 2-connected chordal graph and $H_{\rm r}$ is a cycle of length at least 4, then $(H_{\rm b}, H_{\rm r})$ is highly Ramsey-infinite.

Our proofs are mostly constructive, and consist of two parts- constructing *determiners and senders*, and extending a construction from [18]. We talk now a little about these two parts.

1.3. **Determiners and Senders.** Graphs called *senders* and/or *determiners* were used in [5, 6, 9] and [8] to show that various pairs were Ramsey-infinite. The study of senders and determiners was extended in [10], where they used them to find Ramsey-minimal graphs with various properties, including the stronger notion of Ramsey-infinite mentioned above, which was a precursor to the notion of highly Ramsey-infinite. They were also used in [16, 17] and [18].

Given graphs $H_{\rm b}$ and $H_{\rm r}$, a (negative) *sender* for $(H_{\rm b}, H_{\rm r})$ is a graph S with two *signal edges*, such that under any good colouring, the signal edges get different colours. Given senders, it is easy to construct infinitely many graphs that are Ramsey for $(H_{\rm b}, H_{\rm r})$; we simply take any odd number of them, and string them together in an odd cycle by identifying their signal edges.

A *blue (resp. red) determiner* for $(H_{\rm b}, H_{\rm r})$ is a graph D with a signal edge that is necessarily blue (resp. red) under any good colouring of D. Clearly determiners cannot exist if $H_{\rm b} = H_{\rm r}$, but when they exist they can often be used to construct senders.

For 3-connected graphs $H_{\rm b}$ and $H_{\rm r}$, when we identify signal edges of two senders, the 3-connectedness ensures that there are no copies of $H_{\rm b}$ or $H_{\rm r}$ that were not already in one of the senders. So taking our senders to be minimal, the above construction gives infinitely many Ramsey-minimal graphs. When $H_{\rm b}$ and $H_{\rm r}$ are not 3-connected, it takes more work to control copies of $H_{\rm b}$ and $H_{\rm r}$ that are created when we put even two senders together.

Indeed, in [2], Burr defined a subclass of the 2-connected graphs and showed that if both $H_{\rm b}$ and $H_{\rm r}$ were from this class, then either $(H_{\rm b}, H_{\rm r})$ is Ramsey-infinite, or there exists a determiner. The idea is that if there is a determiner, one should be able to use it to show that $(H_{\rm b}, H_{\rm r})$ is Ramsey-infinite. The determiners in this case, however, were not sufficient to this task. We talk more of this in Subsection 3.4.

In [17], in the case that $H_{\rm b} = H_{\rm r}$ is a cycle, we were able to construct senders that were sufficient for to showing $(H_{\rm b}, H_{\rm r})$ to be Ramsey-infinite by ensuring that they had the same girth as $H_{\rm b}$.

In Section 3 of this paper we continue the study of determiners and senders. We define a *safe sender*, whose existence automatically ensures that $(H_{\rm b}, H_{\rm r})$ is Ramsey-infinite. We focus on pairs of two connected graphs, and show that for many pairs, the existence of

a *safe determiner* implies the existence of a safe sender. Using [1], we then construct a safe determiner for $(H_\mathrm{b}, H_\mathrm{r})$ where H_b is a 2-connected chordal graph, and H_r is a cycle of length at least 4. This gives us the safe senders we need for Corollary 5.2 which is mentioned above.

1.4. **Constructing Highly Ramsey-infinite Pairs.** In [16], we used senders (in fact we constructed them for the more general problem in which k colours are used) to show that pairs of cliques are highly Ramsey-infinite. In [17], a similar but more complicated construction was used for $H_\mathrm{b} = H_\mathrm{r}$ being an odd cycle of girth at least 7. The construction was complicated by the fact that it was done for k colours, while in [16] we were able to deal with more colours by a simple induction. While we were unable to do this construction for five cycles, in [18] we used a much simplified version of the construction from [16] to deal with the two colour version for five cycles.

Using the simplified construction from [18], we are now able to deal with a more general class of graphs. The construction depends on the existence of safe senders, but as it adds little complication, we do it also for many pairs of graphs for which safe senders are not known to exist. Specifically, we show that if H_b is non-bipartite, and H_r is

- a cycle of length at least 4 (Theorem 5.1), or
- a non-complete 3-connected graph (Theorem 7.1),

then $(H_\mathrm{b}, H_\mathrm{r})$ is highly Ramsey-infinite if there exists a safe sender.

We modularise the construction so that it is easier to do the different cases, and so that not so much work has to be repeated if other cases are done later. The first part of the construction is a gadget $\mathscr{G}$ which we construct in Section 4 for any pair of graphs for which safe senders exist. The second part, depends on the particular graphs H_b and H_r, and is done twice, once in Section 5 to prove Theorem 5.1 , and once in Section 7 to prove Theorem 7.1.

2. Definitions and Notation

We associate a graph G with its edge-set. Thus a subset S of the edges of G is considered as a subgraph of G. In this case, the vertex set of S is taken to be the set of vertices of G that are incident with some edge of S. Therefore $G \setminus S$ denotes the subgraph of G constructed by removing all edges of S, and any vertices left independent.

For a graph G and a subset V of its vertex set $V(G)$, $G|_V$ is the subgraph induced by the vertices in V, and $G \setminus V = G|_{V(G)\setminus V}$. We will always denote the edge between vertices u and v by uv. So $G \setminus \{u,v\}$ is the graph we get by removing the vertices u and v from G, while we get $G \setminus \{uv\}$ by removing only the edge uv.

By a colouring ϕ of a graph G, we mean a function $\phi : G \to \{b,r\}$ from the edge-set of G to the two element set $\{b,r\}$. For any subgraph S of G, $\phi(S)$ is the set of colours assigned to edges in S. We write $\phi(S) = b$ or $\phi(S) = r$ for $\phi(S) = \{b\}$ or $\phi(S) = \{r\}$ respectively, and say ϕ is *blue* on S or ϕ is *red* on S respectively. If the colouring is understood, we simply say S is blue or red.

For graphs H and G, *a copy of H in G* is a subgraph of G isomorphic to H. It need not be an induced subgraph of G. Where H_b and H_r are understood, we say a colouring ϕ of a graph G is *good* if there is no blue copy of H_b or red copy of H_r in G.

In this paper we will frequently construct graphs from others by identifying edges or vertices. It will thus be useful to introduce some notation.

Given a graph G and vertices v and v' of G, we say we identify v and v' to mean we add an edge between them if it does not exist, and then contract it. We denote the new

graph by $G/\{v=v'\}$, and refer to the contracted vertex as either v or v'. More generally, an *identification* is the symbol '$v=v'$' for vertices v and v' of G. Given a set $\mathscr{I}=\{v_i=v_i' \mid i=1,\ldots,d\}$ of identifications, the graph $G/\mathscr{I}$ is the graph constructed from G by identifying the vertices v and v' for each identification $v=v'$ in $\mathscr{I}$.

Usually when we identify vertices in G, it is the union $G_1\cup G_2$ of disjoint graphs G_1 and G_2, in which case, we write $G_1\vee_{\mathscr{I}} G_2$ for $(G_1\cup G_2)/\mathscr{I}$. As a series of contractions can be done in any order this is clearly associative:

$$(G_1\vee_{\mathscr{I}_1} G_2)\vee_{\mathscr{I}_2} G_3 = G_1\vee_{\mathscr{I}_1}(G_2\vee_{\mathscr{I}_2} G_3).$$

So we write $G_1\vee_{\mathscr{I}_1} G_2\vee_{\mathscr{I}_2} G_3$.

Often we will identify edges $e=uv$ and $e'=u'v'$, and will not care about in which of the two possible ways we do this. In this situation we write $e=e'$ in $\mathscr{I}$ to mean $u=u'$ and $v=v'$.

If ϕ is a colouring of G and ϕ' is a colouring of G' such that for any $u=u'$ and $v=v'$ in $\mathscr{I}$ with $uv\in G$ and $u'v'\in G'$ we have $\phi(uv)=\phi'(u'v')$, then $\phi\vee_{\mathscr{I}}\phi'$ is the unique colouring of $G\vee_{\mathscr{I}} G'$ that restricts on G to ϕ and on G' to ϕ'. We also say that $\phi\vee_{\mathscr{I}}\phi'$ is the *extension of* ϕ *by* ϕ' (or of ϕ' by ϕ).

The following equivalent definition of *highly Ramsey-infinite* which was used in [18], makes our proofs easier to read.

Lemma 2.1. *A pair* $(H_{\mathrm{b}},H_{\mathrm{r}})$ *is* highly Ramsey-infinite *if there is some constant* c *such that for all* $m\geq 3$ *there are* 2^{m^2} *different labelled graphs on at most* $c\cdot m$ *vertices that are Ramsey-minimal for* $(H_{\mathrm{b}},H_{\mathrm{r}})$.

3. Determiners and Senders

Let H_{b} and H_{r} be fixed connected graphs.

Definition 3.1. Let F be a graph and $S\subset F$ be a subgraph. A good colouring ϕ_F of F is *safe* at S if for any graph G, and set $\mathscr{I}=\{s=s' \mid s\in S\}$ identifying each edge s in S with some edge s' in G, the following is true. The colouring $\phi_F\vee_{\mathscr{I}}\phi_G$ is a good colouring of $F\vee_{\mathscr{I}} G$ if and only if ϕ_G is a good colouring of G. We say a colouring is safe at an edge s if it is safe at $\{s\}$.

Observe the following:

(i) A good colouring ϕ of G being safe on edges s and t does not imply that ϕ is good on the set $\{s,t\}$, however, if s and t far enough apart (more than the diameter of H_{b} and H_{r}), or in different components of G, then it does.

(ii) The construction $F\vee_{\mathscr{I}} G$ may identify vertices and edges of S, this is accounted for in the definition.

(iii) A good colouring of F is necessarily safe at $S=F$, as for any G and $\mathscr{I}$ satisfying the requirements of definition, $G=F\vee_{\mathscr{I}} G$. Similarly, a good colouring of F is safe at any union of components of F.

Often in our constructions we will start with a graph G that we want to show to be Ramsey-minimal. We take a gadget F having only one good colouring ϕ, and attach it to G at a common subgraph S on which F is safe. This makes the graph $G'=F\vee_{\mathscr{I}} G$ (for some set $\mathscr{I}$ of identifications identifying the common subgraph S). We can then say that the only good colourings of G' when restricted to G must agree with ϕ on S. This allows us some control over the colourings of G, which we can now work with, effectively ignoring F.

The following lemma, which follows immediately from the associativity of the above construction allows us to 'maintain safeness' as we construct graphs.

Lemma 3.2. *Let ϕ_F be a good colouring of a graph F which is safe at S. Let G be a graph, and let $\mathscr{I} = \{s = s' \mid s \in S\}$ identify each edge s of S with an edge s' of G. Let ϕ_G be a good colouring of G that is safe at $S_G \subset G$ and satisfies $\phi_G(s') = \phi_F(s)$ for all $s \in S$. Then the good colouring $\phi_F \vee_{\mathscr{I}} \phi_G$ of $F \vee_{\mathscr{I}} G$ is safe at S_G.*

Proof. Assume that $\phi_F \vee \phi_G$ is not safe at S_G. Then there is some graph G', some set of identifications $\mathscr{I}'$ of $S_G \cup G'$, and some good colouring $\phi_{G'}$ of G' such that $(\phi_F \vee_{\mathscr{I}} \phi_G) \vee_{\mathscr{I}'} \phi_{G'}$ is well defined, but not a good colouring of $(F \vee_{\mathscr{I}} G) \vee_{\mathscr{I}'} G'$. By associativity this is equal to the colouring $\phi_F \vee_{\mathscr{I}} (\phi_G \vee_{\mathscr{I}'} \phi_{G'})$ which is therefore not a good colouring of $F \vee_{\mathscr{I}} (G \vee_{\mathscr{I}'} G')$. However, as ϕ_G is safe at S_G and agrees with ϕ_H on $S_G \subset \mathscr{I}'$ (or else $(\phi_F \vee_{\mathscr{I}} \phi_G) \vee_{\mathscr{I}'} \phi_{G'}$ would not be well defined), we have that $\phi_G \vee_{\mathscr{I}'} \phi_{G'}$ is good. This contradicts the fact that ϕ_F is safe on $S_G \subset I$. □

Now we define our gadgets.

Definition 3.3. A *safe blue determiner* (D_b, s) consists of a graph D_b and a edge $s \in D_b$, called a *signal edge*, which satisfy the following properties.

(i) All good colourings ϕ of D_b have $\phi(s) = b$.
(ii) There exists a good colouring of D_b that is safe at s.

A *safe red determiner* (D_r, s) is defined similarly.

Given a graph G with an edge e we say we 'attach a safe blue determiner at e' to mean for a safe blue determiner (D_b, s) disjoint from G we construct the graph $D_b \vee_{\{s=e\}} G$. We say the same of safe red determiners.

Definition 3.4. A *safe (negative) sender* (S, s, t) consists of a graph S with edges $s, t \in S$ satisfying the following properties.

(i) Under any good colouring ϕ of S, $\phi(s) \neq \phi(t)$.
(ii) There exist good colourings ϕ_1 and ϕ_2 of S, safe at $\{s, t\}$, such that $\phi_1(s) = b$ and $\phi_2(s) = r$.

A *safe positive sender* (S, s, t) consists of a graph S with edges $s, t \in S$ satisfying the above properties, except in property (i) it has '$\phi(s) = \phi(t)$' instead of '$\phi(s) \neq \phi(t)$'.

Now given a graph G with edges e and f, we say we 'connect e and f with a negative (positive) sender' to mean we construct the graph $G \vee S/\{e = s, f = t\}$, where (S, s, t) is a safe negative (resp. positive) sender disjoint from G.

The following Lemma is immediate.

Lemma 3.5. *Let (H_b, H_r) be an asymmetric pair for which there exists a safe (negative) sender. Then there exists safe positive sender, as well as safe red and blue determiners.*

Proof. Assuming the existence of a safe negative sender we construct a safe positive sender (S, s, t) from a set of three disjoint edges s, m and t as follows. Connect s and m with a negative sender S_1 and connect m and t with a negative sender S_2. Property (i) of the definition of a safe positive sender is easily verified. For property (ii), the existence of a colouring ϕ_1 with $\phi_1(s) = b$ follows from the existence of a colouring of S_1 that is red on s and blue on m, and the existence of a colouring of S_2 that is blue on m. To see that this colouring is safe at $\{s, t\}$ observe that its restriction to the colouring ϕ of $\{s, m, t\}$ is safe on $\{s, t\}$ by comment (iii) following Defintion 3.1. If follows by Lemma 3.2 that ϕ_1 is safe on $\{s, t\}$. The existence of a safe colouring ϕ_2 is similar.

To construct a safe red determiner (D_r, s) we proceed as follows, first assuming that H_r contains no copy of H_b. From a copy of H_r and an edge s construct D_r by connecting e and s by a safe negative sender S_e for every edge e of H_r. If ϕ is a good colouring of D_r then the edge s must be red, or else every edge of H_r would be red, which is impossible under a good colouring. On the other hand, the colouring ϕ_1 of $H_r \cup e$ defined by $\phi_1(e) = b$ if $e \in H_r$, and $\phi_1(s) = r$ is good by the assumption that H_r contains no copy of H_b. By property (ii) of the safe negative sender S_e (for each e), there is a colouring ϕ_e of S_e that agrees with ϕ_1 on e and s and is safe at s. Extending ϕ_1 by each of these ϕ_e and applying Lemma 3.2 each time, we get a colouring ϕ that is safe at s. Thus D_r is a safe red determiner.

If H_r contains a copy of H_b then H_b cannot contain a copy of H_r and we can do the same construction using safe positive senders in place of the safe negative senders.

The construction of safe blue determiners is the same. □

Starting with a safe positive sender, it is a simple exercise to construct safe red and blue determiners, and with these three gadgets, a safe negative sender. So one could prove a similar Lemma starting with a positive sender. One can also construct a negative sender with just a single safe determiner, but assuring that this sender is safe is non-trivial. In the rest of this section, we do manage to construct safe negative senders for certain pairs of graphs.

3.1. **Safeness by 2-blocks.** The definitions of safe determiners and senders are formulated so that they are easily used. In this section, we reformulate the definitions in a way that helps in constructing them, especially for 2-connected graphs.

We start by elaborating on some of our earlier definitions. First we give another definition of a copy of a graph H in another graph G. An *embedding* i of H into G is an injective vertex mapping $i : V(H) \to V(G)$, such that $i(u)i(v)$ is in G if uv is in H. Let $i(H)$ denote the not necessarily induced subgraph of G on $i(V(H))$ such that $i(u)i(v)$ is in $i(H)$ if and only if uv is in H. We call $i(H)$ *a copy of H in G*.

Where H_b and H_r are understood, a colouring ϕ of a graph G is *blue-good* if there is no blue copy of H_b in G, and *red-good* if there is no red copy of H_r. Clearly, a colouring is good if and only if it is red-good and blue-good.

Analogous to Definition 3.1, we can define a colouring that is red-safe or blue-safe on a set S, but we will need it only for single edge sets.

Definition 3.6. Let F be a graph and s an edge in F. A good colouring ϕ_F of F is *blue-safe* at s if for any graph G, edge $e \in G$, and colouring ϕ_G of G with $\phi_G(e) = \phi_F(s)$, the following is true. The colouring $\phi_F \vee_{\{s=e\}} \phi_G$ is a blue-good colouring of $F \vee_{\{s=e\}} G$ if and only if ϕ_G is a blue-good colouring of G. *Red-safe* is defined analogously.

Remark 3.7. It is clear that a colouring is safe at s if it is blue-safe and red-safe at s. Moreover, Lemma 3.2 is clearly true for 'red-safe' or 'blue-safe' in place of 'safe'.

We now work towards a structural definition of what it means for a colouring to be safe at an edge s. We need a way to talk about how new copies of H_b or H_r can 'slip through' a signal edge.

Definition 3.8. Given a 2-connected graph H, a 2-*block* (B, u, v) of H consists of a proper induced subgraph B of H and vertices u and v of B such that $B \setminus \{u, v\}$ is a component of $H \setminus \{u, v\}$. The set $\{u, v\}$ is the *boundary* of B, or of (B, u, v). If u and v are adjacent, (B, u, v) is called a *closed* 2-*block*, otherwise it is called an *open* 2-*block*.

Observe that for the cycle C_ℓ the 2-blocks are the paths of lengths 2 up to $\ell - 2$, and all are open. The following should be clear.

Lemma 3.9. *Let F be a graph and $s = xy$ be an edge in F. A colouring ϕ of F with $\phi(s) = r$ is blue-safe at s if and only if there is no blue copy $i(B)$ of an open 2-block (B,u,v) of H_b with $i(u) = x$ and $i(v) = y$. A colouring ϕ of F with $\phi(s) = b$ is blue-safe at s if there is no blue copy $i(B)$ of any* **open or closed** *2-block (B,u,v) of H_b with $i(u) = x$ and $i(v) = y$.*

The same statements hold mutatis mutandis for 'red' in place of 'blue', and together, give a alternate definition of ϕ being safe at s.

Proof. If there is such a blue copy $i(B)$ in F, with $\phi(s) = r$, then where B' is a copy of $H_\mathrm{b} \setminus B$ and $\phi'(B') = b$, ϕ' is a blue-good colouring of B' but $\phi \vee \phi'$ is not a blue-good colouring of

$$F \vee_{\{x=u,y=v\}} B'.$$

Indeed, it contains the blue copy

$$i(B) \vee_{\{x=u,y=v\}} B'$$

of H_b. The case that $\phi(s) = b$ is similar, and the inverse statement in both cases is clear. □

3.2. **Safe Blue Determiners.** The following definition, extending the class $\mathscr{C}(4)$ of 2-connected chordal graphs, is from [1].

Definition 3.10. For integer $\ell \geq 4$, a graph is in the class $\mathscr{C}(\ell)$ if it is 2-connected and has no induced cycle of length ℓ or greater.

In [1], the authors show that for H_r an ℓ-cycle, and H_b in $\mathscr{C}(\ell)$, (H_b, H_r) is Ramsey-infinite. In this subsection we use the proof of their result to show the existence of safe blue determiners for all such pairs.

The following is essentially in the proof of the main result of [1].

Proposition 3.11. *Let H_r be the cycle C_ℓ, and H_b be in $\mathscr{C}(\ell)$. Then there exists a graph G that is Ramsey for (H_b, H_r), and such that the following are true.*

(i) *Every edge of G is in exactly one copy of H_b.*
(ii) *Apart from those in the above mentioned copies of H_b, there are no cycles in G of girth less than ℓ.*

In their proof, they state property (i) explicitly in comment (**) on page 30, and prove it as property (P6) in Section 5.3. They do not state property (ii) explicitly, but it follows from their construction: to construct their graph G, they start with a $|V(H_\mathrm{b})|$-uniform hypergraph $\mathscr{H}$ of girth ℓ, and replace every hyperedge with a copy of H_b.

Using this result, we show the following.

Corollary 3.12. *Let H_r be the cycle C_ℓ, and H_b be in $\mathscr{C}(\ell)$. Then there exists a safe blue determiner for $(H_\mathrm{b}, H_\mathrm{r})$.*

Proof. Let G be the graph from Proposition 3.11. As any Ramsey-minimal subgraph of G still has properties (i) and (ii), we may assume that G itself is Ramsey-minimal.

Now for some edge e of G let H^* be the copy of H_b in G containing e. Then (G', e) is a blue determiner, where

$$G' = G \setminus (H^* \setminus \{e\}).$$

Indeed, as G is Ramsey-minimal, G' has a good colouring ϕ. Any such ϕ must be blue on e or we can extend it to a good colouring of G by letting it be blue on $H^* \setminus \{e\}$. To see that such ϕ is safe at e observe that from properties (i) and (ii) of G from Proposition 3.11, e is in no cycle of length shorter than ℓ. However, any 2-block of H_b or H_r has a path of length at most $\ell - 2$ between its boundary vertices (excluding the edge e itself if

the 2-block is closed), so e cannot be the boundary of a 2-block. This means that ϕ is safe at e. □

The obvious approach to make red determiners now is to take a copy of H_b and attach a safe blue determiner to every edge but some edge s. Indeed, this gives a red determiner with signal edge s. This determiner is red-safe, but is not, in general, safe. We must be careful about how we choose s.

3.3. **From Safe Determiners to Safe Senders.** In this section, we observe that for a safe sender to exist for a pair $(H_\mathrm{b}, H_\mathrm{r})$ of 2-connected graphs, both H_b and H_r must have what is called a *safe edge*. We show that if they do both have safe edges, and there is a safe blue (or red) determiner, then there are safe blue and red determiners and a safe sender.

All graphs in $\mathscr{C}(4)$ have safe edges, so with the results from the previous section, this will give us safe senders for any pairs (H_b, H_r) with $H_b \in \mathscr{C}(\ell)$ and $H_b = C_\ell$.

Definition 3.13. In a two connected graph H, an edge $e = xy$ is *safe* if there is no copy $i(B)$ in H of any *open* 2-block (B, u, v) of H, with $i(u) = x$ and $i(v) = y$.

If H_b has no safe edge, then there can be no blue-safe red determiner for $(H_\mathrm{b}, H_\mathrm{r})$ and so no safe red determiner. Indeed, starting from any good colouring of a red determiner (R, s) if we re-colour s blue, then it lies in a blue copy of H_b. That is $s = i(e_s)$ for some edge e_s of H_b and some blue copy $i(H_\mathrm{b})$ of H_b in R. But e_s is not safe so there is some copy $j(B)$ of an open 2-block (B, u, v) of H_b in H_r with $j(u) = x$ and $j(v) = y$. Then $i \circ j(B)$ is blue copy of (B, u, v) in R with $i \circ j(u)$ and $i \circ j(v)$ being the endpoints of s.

Similarly, for there to be a safe sender for $(H_\mathrm{b}, H_\mathrm{r})$ both H_b and H_r must have safe edges. If H is a cycle, or a 2-connected graph in which all 2-blocks are closed (such as graphs in $\mathscr{C}(4)$), then every edge is safe.

Now given a safe blue determiner for 2-connected H_b and H_r, we have that H_r must have a safe edge. We show that if H_b also has a safe edge, then there is a safe blue determiner. We then go on to show there is a safe sender.

Lemma 3.14. *Let H_b be 2-connected, and contain a safe edge s. Let H_r be such that there is a safe blue determiner for $(H_\mathrm{b}, H_\mathrm{r})$. Then there is a safe red determiner for $(H_\mathrm{b}, H_\mathrm{r})$.*

Proof. Let the edges of $H_\mathrm{b} \setminus \{s\}$ be labelled $e_1, \dots, e_m$. Construct D from H_b by attaching a safe blue determiner to every edge but s. That is, for $i = 1, \dots m$, let (B^i, t^i) be a copy of a safe blue determiner (B, t) and let D be $(\bigcup_{i=1}^m B^i) \vee_{\mathscr{I}} H_\mathrm{b}$ where $\mathscr{I} = \{e_i = t^i \mid i = 1, \dots, m\}$.

It is clear that any good colouring of D is red on s, as the copies of (B, t) ensure that it is blue on every other edge of H_b. We now show that there is a good colouring that is safe on s. Indeed, let ϕ be a good colouring of $\bigcup_{i=1}^m B^i$ that is safe (and necessarily blue) on on each of the t^i. As any two t^i are in different components of $\bigcup_{i=1}^m B^i$, ϕ is safe at $\{t_1, \dots t_m\}$. Let ϕ' be the colouring on H_b that is blue on every edge but s. Observe that ϕ' is a good colouring of H_b and safe at s: it is clearly red-safe at s, and is blue-safe at s because s is a safe edge of H_b.

Now, the colouring $\phi \vee_{\mathscr{I}} \phi'$ is a good colouring of D because ϕ is safe at s and ϕ' is a good colouring of H_b. Further, we see that it is safe at s by constructing $(\bigcup_{i=1}^m B^i) \vee_{\mathscr{I}} H_\mathrm{b}$ as

$$B^m \vee_{\{e_m = t_m\}} (\dots B^2 \vee_{\{e_2 = t_2\}} (B^1 \vee_{\{e_1 = t_1\}} H_\mathrm{b}) \dots))$$

attaching one blue determiner at a time, and applying Lemma 3.2 each time. □

Before we construct the sender, we construct a preliminary gadget.

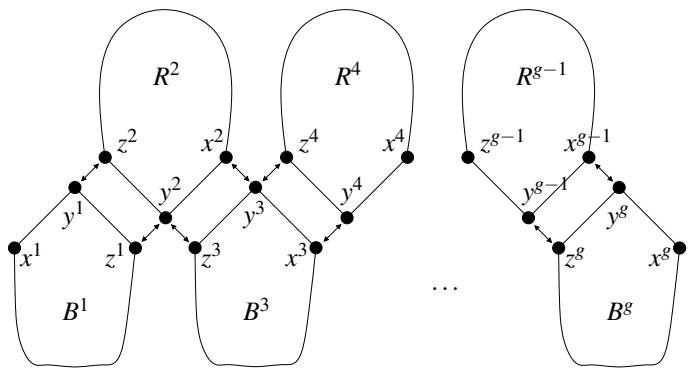

FIGURE 1. The construction of D_g from copies B^i of H_b and copies R^i of H_r.

Lemma 3.15. *Let H_b and H_r be 2-connected graphs and such that safe blue and red determiners exist for $(H_\mathrm{b}, H_\mathrm{r})$. Then there exists a* safe not-both-blue *gadget (N_b, s, t), that is, a graph N_b containing edges s and t such that*

(i) *Under any good colouring ϕ of N_b, $\phi(\{s,t\}) \neq b$.*
(ii) *There exist good colourings ϕ_1 and ϕ_2 of N_b, safe at $\{s,t\}$, such that $\phi_1(s) = r$ and $\phi_2(t) = r$.*

Similarly, there is a safe not-both-red *gadget, which is defined by switching the roles of b and r.*

Proof. As there are safe red and blue determiners, H_b and H_r have safe edges xy and $x'y'$ respectively. Let z be a neighbour of y in H_b and z' a neighbour of y' in H_r.

For each positive odd integer i let B^i be a copy of H_b and let x^i, y^i and z^i be the copies of x, y, and z in B^i. For each even i, let R^i be a copy of H_r and let x^i, y^i and z^i be the copies of x, y, and z in R^i.

Where $\mathscr{I}_1 = \{y^1 = z^2, z^1 = y^2\}$, and $\mathscr{I}_i = \{y^i = z^{i+1}, x^i = y^{i+1}\}$ for $i \geq 2$, let

$$D_g = B^1 \vee_{\mathscr{I}_1} R^2 \vee_{\mathscr{I}_2} B^3 \vee_{\mathscr{I}_3} \cdots \vee_{\mathscr{I}_{g-1}} B^g$$

for odd $g \geq 3$. See Figure 1.

Let P be the even path $\{x^1y^1, y^1z^1 = z^2y^2, y^2x^2 = z^3y^3, \ldots, y^{g-1}x^{g-1} = z^gy^g, y^gx^g\}$ in D_g, and let $s = x^1y^1$ and $t = y^gx^g$ be the ends of P. Let ϕ be the colouring of $D_g \setminus P$ that is blue on $(B^1 \cup B^3 \cup \cdots \cup B^g) \setminus P$ and red on $(R^2 \cup R^4 \cup \cdots \cup R^{g-1}) \setminus P$.

Any extension of ϕ to a good colouring of D_g must have $\phi(s,t) \neq \{b\}$. Indeed, if for such ϕ we have $\phi(x^1y^1) = b$ then we must have $\phi(z^2y^2) = r$ or ϕ is blue on B^1. Continuing in this way, we show that ϕ alternates on the path P, so is red on $t = y^gx^g$. Similarly, we can show that if $\phi(y^gx^g) = b$, then $\phi(x^1y^1) = r$.

Now let ϕ' be the extension of ϕ to D_g that is blue on $s = x^1y^1$ and altenates on P. We claim that this is a good colouring of D_g, that is safe at $s = x^1y^1$ and $t = y^gx^g$.

Indeed, the subgraph D_b of D_g of edges that are blue under ϕ' is

$$D_b = (B^1 \setminus \{y^1z^1\}) \vee_{\{z^1=z^3\}} (B^3 \setminus \{y^3x^3\}) \vee_{\{x^3=z^5\}} \cdots \vee_{\{x^{g-2}=z^g\}} (B^g \setminus \{y^gx^g\}).$$

The maximal 2-connected subgraphs of D_b are copies of H_b less an edge, so D_b cannot contain a copy of H_b. Similarly, the 2-connected subgraphs of the graph D_r of edges that are red under ϕ' are proper subgraphs of H_r. So ϕ' is a good colouring of D_g.

We now verify that ϕ' is safe at $s = x^1y^1$. It is red-safe at s because all edges incident to x^1 are blue, so there is no red copy $i(B)$ of a 2-block (B,u,v) of H_r in D_g with $i(u) = x^1$. It is blue safe at s because any blue copy $i(B)$ of a 2-block (B,u,v) of H_b in D_g with $i(u) = x^1$ and $i(v) = y^1$ must be contained in the maximal 2-connected subgraph $B^1 \setminus \{y^1z^1\}$ of D_b. However, as x^1y^1 is a safe edge of $B^1 = H_\mathrm{b}$, there is no such 2-block of H_b in B^1. The proof that ϕ' is safe at t is the same.

Similarly, the extension of ϕ to D_g that is red on S and alternates on P is a good colouring that is safe on s and t. Taking g large enough, both of these extensions of ϕ are safe on $\{s,t\}$.

Now attaching a safe red determiner to every edge on which ϕ is red, and a blue determiner to every edge on which it is blue, the resulting graph N_b has the required properties by Lemma 3.2.

□

Corollary 3.16. *Let H_b and H_r be 2-connected graphs and such that safe blue and red determiners exist for $(H_\mathrm{b}, H_\mathrm{r})$. Then there is a safe negative sender for $(H_\mathrm{b}, H_\mathrm{r})$.*

Proof. Take a safe not-both-blue gadget (N_b, s_b, t_b) from Lemma 3.15 and a safe not-both-red gadget (N_r, s_r, t_r). Then the graph $N_b \vee_{\{s_b=s_r, t_b=t_r\}} N_r$ is a safe negative sender. □

Corollary 3.17. *Let H_b be a 2-connected chordal graph, and H_r be a cycle of girth $\ell \geq 4$, then there is a safe negative sender for $(H_\mathrm{b}, H_\mathrm{r})$.*

Proof. As H_b is in $\mathscr{C}(4) \subset \mathscr{C}(\ell)$, Corollary 3.12 gives us a safe blue sender for $(H_\mathrm{b}, H_\mathrm{r})$. In chordal graphs, all edges are safe so Lemma 3.14 gives us a safe red sender. Thus by Corollary 3.16 there is a safe negative sender for $(H_\mathrm{b}, H_\mathrm{r})$. □

3.4. **Remarks on Determiners and Senders.** In [2] the class of 2-connected graphs G in which every 2-block is closed was considered. It was shown that if both H_b and H_r are in this class, then either $(H_\mathrm{b}, H_\mathrm{r})$ is Ramsey-infinite, or there exists either a blue or a red determiner. The blue determiners in this case are automatically *red-safe* but are not safe. Further, every edge in such graphs G are safe, so if we could construct safe determiners from these red-safe blue determiners, we would have that $(H_\mathrm{b}, H_\mathrm{r})$ is Ramsey-infinite. This would be interesting.

The set $\mathscr{C}(4)$ of chordal 2-connected graphs have the property that every 2-block is closed, so every edge is safe. It is not hard to come up with a graph in $\mathscr{C}(6)$ that has no safe edges. So there are pairs $(H_\mathrm{b}, H_\mathrm{r})$ that are Ramsey-infinite by [1] for which there are no safe senders. However, one could extend Definition 3.1 to sets Φ of colourings of F, saying the set is safe at S if any colouring ϕ_G of G is good if and only if $\phi \vee_{\mathscr{I}} \phi_G$ is good for some $\phi \in \Phi$. We could then replace the good colouring in (ii) of Definition 3.3 with a set of good colourings. Determiners according to this less restrictive definition would be just as useful, and may well exist for graphs H_b and H_r not having safe edges. We refrain from this more complicated definition though as we have no examples of pairs (H_b, H_r) for which these more general determiners/senders exist, while ours do not. Finding such a pair would be interesting. Could these more general safe determiners and senders exist for every pair of 2-connected graphs?

In [1] they say

> ... our proof [that (H_b, H_r) is Ramsey-infinite for $H_b \in \mathscr{C}(\ell)$ and $H_r = C_\ell$] is probabilistic. It would be interesting to prove this result by explicit constructions.

In the case that $\ell = 4$, this section reduces their result to Proposition 3.11, which is the only non-constructive result we use in this paper (apart from Lemma 6.1, which could be replaced with a constructive result from [12]). In fact, with similar constructions to those in this section we could do this for all ℓ: from the safe blue determiner in Corollary 3.12, one could construct a safe not-both-red gadget and a red-safe not-both-blue gadget, and from that a red-safe sender. With these three gadgets, it is not hard to show that (H_b, H_r) is Ramsey-infinite. We do not include these constructions, as our highly Ramsey-infinite construction does not work with them. So the result of [1] can be reduced to Proposition 3.11. As such, we feel a constructive proof Proposition 3.11 would be interesting, even in the case that H_b and H_r are both cycles.

4. Other Gadgets

In this section, we assume that $(H_\mathrm{b}, H_\mathrm{r})$ is an asymmetric pair of graphs for which there are safe senders (and so safe determiners), and construct a gadget $\mathscr{G}$ which we will use in Sections 5 and 7. We modularise our construction into a couple of lemmas to simplify it and to make later referencing easier.

The assumption that $(H_\mathrm{b}, H_\mathrm{r})$ is asymmetric can be relaxed by replacing determiners with a simple construction of senders. This is discussed at the end of Section 7.

4.1. **Preliminary Gadgets.** It is clear that for any graph P and good colouring p of P we can construct a graph G containing P such that any good colouring of G restricts on P to p, and such that there is a such a good colouring that is safe on P. Indeed, we just attach the appropriate determiner to each edge of P. Using senders as well as determiners, it is not hard to extend this to a similar statement that allows exactly two different good colourings on P. We do this now by constructing the *two-colourings gadget.* $T = T(P, p_0, p_1)$.

Lemma 4.1. *Let $(H_\mathrm{b}, H_\mathrm{r})$ be an asymmetric pair for which there exists a safe sender. Then given a graph P with two good colourings p_0 and p_1, there exists a graph $T = T(P, p_0, p_1)$ containing P and an another edge s, which satisfies the following.*

(i) *For any good colouring ϕ of T, ϕ restricts on P to p_0 if $\phi(s) = r$ and restricts on P to p_1 if $\phi(s) = b$.*
(ii) *There exist good colourings ϕ_r and ϕ_b of T with $\phi_r(s) = r$ and $\phi_b(s) = b$ that are safe on $P \cup \{s\}$.*

Proof. Start with the disjoint graphs P and s. Any good colouring is safe on $P \cup \{s\}$. Construct T from $P \cup \{s\}$ by doing the following for every edge e in P. (Recall that by Lemma 3.5, we have safe positive senders and safe red and blue determiners.)

- If $p_0(e) = p_1(e) = b$, then attach a safe blue determiner to e.
- If $p_0(e) = p_1(e) = r$, then attach a safe red determiner to e.
- If $p_0(e) = r$ and $p_1(e) = b$, then connect e and s with a safe positive sender.
- If $p_0(e) = b$ and $p_1(e) = r$, then connect e and s with a safe negative sender.

For each step of the construction, Lemma 3.2 ensures that the good colourings of the constructed graph which remain good, remain safe on $P \cup \{s\}$. The verification of properties (i) and (ii) is otherwise straightforward. □

Our goal is to extend this to a similar statement, but which allows the two fixed colourings to vary under permutation of certain subgraphs. The first step in this extension is the following *one-in-m gadget* $O(m)$.

Lemma 4.2. *Let $(H_\mathrm{b}, H_\mathrm{r})$ be an asymmetric pair (with $|H_\mathrm{r}| \geq 3$) for which there exists a safe sender. Then there exists a constant c' such that for all $m \geq 2$ there exists a graph $O = O(m)$ with a subset $S = \{s_1, \ldots, s_m\} \subset O$ satisfying the following properties.*

(i) *Under any good colouring ϕ of O, there exists α in $[m]$ such that $\phi(s_\alpha) = b$.*
(ii) *For any choice of α in $[m]$, there exists a good colouring ϕ of O which is safe on S, and is red on every edge of S except s_α.*
(iii) *O has less than $c' \cdot m$ vertices.*

Proof. The proof is by induction on m. First assume that $m \leq |H_\mathrm{r}| =: h$. Let O' be a copy of H_r with a red determiner attached to every edge but m of them: $s'_1, \ldots, s'_m$. Observe that O', and $s'_1, \ldots, s'_m$ satisfy property (i), and satisfy (ii) except that the good colouring ϕ may not be safe on $\{s'_1, \ldots, s'_m\}$. Let $s_1, \ldots, s_m$ be new disjoint edges, and connect s'_i to s_i with a copy of S^+ for each $i = 1, \ldots, m$. The resulting graph O has all the necessary properties.

Now let $m > h$, and assume that $O(m-h+2)$ with signal edges $e_1, \ldots, e_{m-h+2}$ and $O(h)$ with signal edges $f_1, \ldots, f_h$ exist. Construct $O = O(m)$ by connecting e_1 and f_1 with a negative sender, and let $S = \{e_2, \ldots, e_{m-h+2}\} \cup \{f_2, \ldots, f_h\}$. One can check that O satisfies properties (i) and (ii).

For (iii) we observe that $O(m)$ is made up of the graph $O(i)$, for some $i \leq h$, and at most $4(m-h)$ copies of S^+. Indeed, this is clearly true if $m \leq h$. If $h > m$ then we add $h-2$ signal edges in every step of the induction by adding a copy of S^+ and a copy of $O(h)$, which consists of h copies of S^+. As $(h+1)/(h-2) \leq 4$, $O(m)$ consists of at most one copy of $O(i)$ and $4(m-h)$ copies of S^+. Thus where $c = max_{i=1}^{h} |V(O(i))|$, $c' = c/m + 4|V(S^+)|$ is sufficient.

□

Putting the two-colouring gadget and the one-in-m gadget together, we build the following gadget that is tailored for our main construction.

4.2. **The Gadget $\mathscr{G} = \mathscr{G}(P, Q, V^*(Q), R, p_0, p_1, q_0, q_1, m)$.** Let graphs P, Q, R, and integer $m \geq 2$ be given. Let $V^*(Q)$ be a subset of $V(Q)$. Let $\mathscr{P}$ be the disjoint union of m copies $P^1, \ldots, P^m$ of P, and $\mathscr{Q}$ be the disjoint union of $2m$ copies $Q^1, \ldots, Q^{2m}$ of Q. Let $V^*(\mathscr{Q}) = \cup_{i \in [2m]} V^*(Q^i)$. Let the graph $S_\mathscr{G} = S_\mathscr{G}(P, Q, V^*(Q), R, m)$ be constructed from the disjoint union of $\mathscr{P}$ and $\mathscr{Q}$ by adding the set $R_\mathscr{Q}$ of all possible edges between R and $V^*(\mathscr{Q})$. (See Figure 3.)

Proposition 4.3. *Let $(H_\mathrm{b}, H_\mathrm{r})$ be an asymmetric pair for which there exists a safe sender. Let P, Q and R be graphs, $V^*(Q)$ be a subset of $V(Q)$, p_0 and p_1 be good colourings of P, and q_0 and q_1 be good colourings of Q. Then there exists a constant c such that for any integer $m \geq 2$ the following is true.*

There exists a graph $\mathscr{G} = \mathscr{G}(P, Q, V^*(Q), R, p_0, p_1, q_0, q_1, m)$ containing $S_\mathscr{G}$, which satisfies the following properties.

(i) *For any good colouring ϕ of $\mathscr{G}$ there exist α and β in $[m]$ such that*
   - *$\phi|_{P^\alpha} = p_1$ and $\phi|_{Q^\beta} = \phi|_{Q^{m+\beta}} = q_1$.*
   - *ϕ is blue on R and $R_\mathscr{Q}$.*

(ii) *For any choice of α and β in $[m]$, there exists a good colouring ϕ of $\mathscr{G}$ that is safe on $S_\mathscr{G}$, and that satisfies the following*
   - *$\phi|_{P^i} = p_0$ for all $i \in [m] \setminus \{\alpha\}$.*
   - *$\phi|_{Q^j} = q_0$ for all $j \in [2m] \setminus \{\beta, m+\beta\}$.*

(iii) *$\mathscr{G}$ has less than $c \cdot m$ vertices.*

Proof. Construct $\mathscr{G}$ from $S_{\mathscr{G}}$ and two one-in-m gadgets O^P and O^Q from Lemma 4.2, as follows.

- For $i = 1, \ldots, m$ let T_P^i be a copy of the two-colourings gadget $T(P, p_0, p_1)$ from Lemma 4.1. Identify the copy of s in T_P^i with the copy of s_i in O^P and identify the copy of P in T_P^i with P^i in $\mathscr{P}$.
- For $i = 1, \ldots, 2m$ let T_Q^i be a copy of the two-colourings gadget $T(Q, q_0, q_1)$ from Lemma 4.1. Identify the copy of s in T_Q^i with the copy of s_i (or s_{i-m} if $i > m$) in O^Q and identify the copy of Q in T_Q^i with Q^i in $\mathscr{Q}$.
- Attach a blue determiner to each edge of R and $R_{\mathscr{Q}}$.

The verification of the properties (i - iii) is immediate from the corresponding properties of the gadgets O and T. The verification that the colourings in (ii) are safe comes from Lemma 3.2 by observing that when restricted to $S_{\mathscr{G}} \cup O^P \cup O^Q$, they are safe on the component $S_{\mathscr{G}}$ (see observation (iii) following Definition 3.1.) □

5. The First Case of our Main Construction

In this section we prove the following.

Theorem 5.1. *Let H_{b} contain an odd cycle and H_{r} be a cycle of girth at least 4. If there exists a safe sender for $(H_{\mathrm{b}}, H_{\mathrm{r}})$, then $(H_{\mathrm{b}}, H_{\mathrm{r}})$ is highly Ramsey-infinite.*

With Corollary 3.16, this gives the following.

Corollary 5.2. *Let H_{b} be chordal 2-connected, and H_{r} be a cycle of girth $\ell \geq 4$, then $(H_{\mathrm{b}}, H_{\mathrm{r}})$ is highly Ramsey-infinite.* □

Let H_{b} and H_{r} be as in the statement of the theorem. We prove the theorem by finding a constant c such that for all integers $m \geq 3$ there are at least 2^{m^2} Ramsey minimal graphs on cm vertices; according to Lemma 2.1, this is enough. We will use the gadget $\mathscr{G}$ of Proposition 4.3, so must define its components. We do this in Subsection 5.1. In Subsection 5.2 we define auxiliary graphs G_0 and G_1, which we use in Subsection 5.3 to define 2^{m^2} different graphs on $V(\mathscr{G})$, containing $\mathscr{G}$, that are Ramsey-minimal for $(H_{\mathrm{b}}, H_{\mathrm{r}})$.

5.1. Definition of $\mathscr{G}(P, Q, R, V^*(Q), p_0, p_1, q_0, q_1, m)$.

5.1.1. *Definition of I_b, m_b, and M_b.* For an independent set I of a graph H, a connected component C of $H \setminus I$ is called *odd* if there is no bipartition of $H|_{V(C)\cup I}$ with I in one of the partite sets. Let $\mathscr{O}_I(H)$ be the set odd components of $H \setminus I$. Let $m_I(H)$ be the maximum size (number of edges) of an element of $\mathscr{O}_I(H)$, and let $M_I(H)$ be the number of elements of $\mathscr{O}_I(H)$ of size $m_I(H)$. (Distinct elements in $\mathscr{O}_I(H)$ may be isomorphic.)

As H_{b} is non-bipartite, $\mathscr{O}_I(H_{\mathrm{b}})$ is non-empty for any independent set I in H_{b}. Over all independent sets of H_{b} let I_b be one that

- minimises $m_b = m_{I_b}(H_{\mathrm{b}})$, and
- subject to this, minimises $M_b = M_I(H_{\mathrm{b}})$.

The following is true.

Claim 5.3. *If H_{b} is a subgraph of H, and I' is an independent set in H, then $m_{I'}(H) \geq m_b$, and if $m_{I'}(H) = m_b$ then $M_{I'}(H) \geq M_b$.*

Proof. Indeed $I = V(H_{\mathrm{b}}) \cap I'$ is an independent set of H_{b} and the odd components of $H_{\mathrm{b}} \setminus I$ are subgraphs of the odd components of $H \setminus I'$, so $m_{I'}(H) \geq m_I(H_{\mathrm{b}}) \geq m_b$ by choice of I_b. If $m_{I'}(H) = m_I(H_{\mathrm{b}}) = m_b$, then distinct maximum elements of $\mathscr{O}_I(H_{\mathrm{b}})$ remain distinct maximum elements of $\mathscr{O}_{I'}(H)$. So $M_{I'}(H) \geq M_I(H_{\mathrm{b}}) \geq M_b$. □

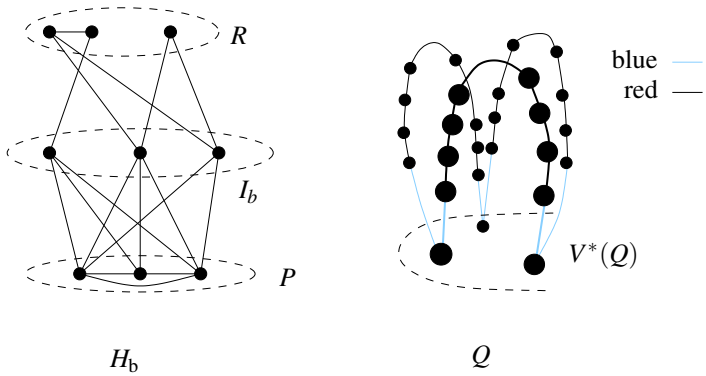

FIGURE 2. P, R, Q and I_b when H_b is as shown and $H_r = C_{11}$. The colouring q_0 of Q is also shown. The set $V^*(Q)$ contains $|I_b| + |P| = 6$ vertices.

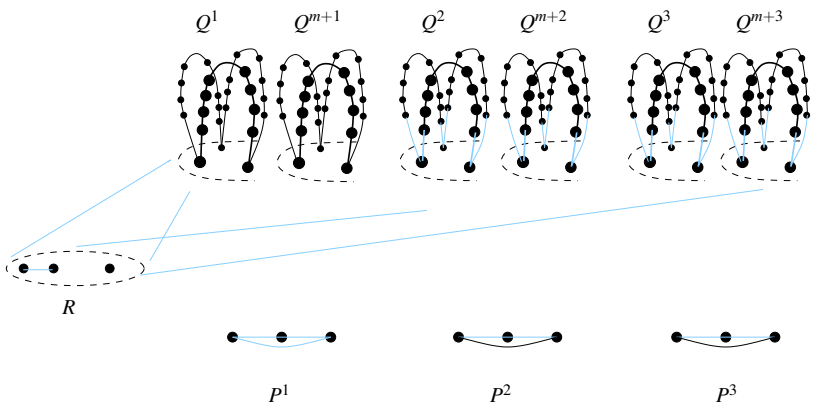

FIGURE 3. A typical colouring of $\mathscr{G}$ restricted to $S_{\mathscr{G}}$ when H_b and Hr are as in Figure 2.

5.1.2. *Definition of P, R, p_0, and p_1.* Let P be a particular odd component of $H_b \setminus I_b$ of size m_b and let R be the graph $H_b \setminus (I_b \cup V(P))$. By choice of I_b, all vertices of P are adjacent to vertices in I_b. Let p_1 be the edge colouring of P that colours every edge blue, and let p_0 be a colouring of P that is red on exactly one edge. See Figure 2 for an example of the partition of a graph H_b into P, I_b and R.

5.1.3. *Definition of Q, q_0 and q_1.* Let $V^*(Q)$ be a set of $|I_b| + |P|$ independent vertices. Construct Q from $V^*(Q)$ by adding a path of length $\ell - 2$ between each pair of vertices in $V^*(Q)$. Let q_1 be the edge colouring of Q that colours every edge red, and let q_0 be a colouring of Q that is blue on the edges incident to vertices of $V^*(Q)$ and red on the other edges. See Figure 2 for an example of Q under the colouring q_0.

5.1.4. *Definition of c, m, and $\mathscr{G}$.* Now that we have defined $P, Q, V^*(Q), R, p_0, p_1, q_0$, and q_1, let c be the constant we get from Proposition 4.3 for these components. Let an integer $m \geq 3$ be fixed, and let $\mathscr{G}$ be the graph $\mathscr{G}(P, Q, V^*(Q), R, p_0, p_1, q_0, q_1, m)$ from Proposition

4.3. Figure 3 shows what the colourings of $\mathscr{G}$ look like, (up to automorphism of the copies of P and Q,) when restricted to the subgraph $S_{\mathscr{G}}$.

5.2. **The auxiliary graphs G_0 and G_1.** Let G_0 be the graph constructed from P and Q by adding all edges between $V(P)$ and $V^*(Q)$. For any copies P' and Q' of P and Q, we say we 'complete P' and Q' to a copy of G_0' to mean we add all edges between $V(P')$ and the copy of $V^*(Q)$ in Q'.

Let G_1 be constructed from G_0 and R by adding the set R_Q of all possible edges between R and $V^*(Q)$ and then attaching blue determiners each edge in R and R_Q. We have the following.

Claim 5.4. *There is no good colouring ϕ of G_1 with $\phi|_P = p_1$ and $\phi|_Q = q_1$.*

Proof. Assume ϕ is such a good colouring of G_1. For any set I of $|I_b|$ vertices in $V^*(Q)$, the subgraph of G_1 induced by $V(P) \cup I \cup V(R)$ contains a copy H of H_b. As $\phi|_P = p_1$ and there are blue determiners on every edge of R and R_Q, ϕ is blue on all edges of H except those between I and $V(P)$. So ϕ must be red on some edge between I and $V(P)$. As $V^*(Q)$ contains $|I_b| + |P|$ vertices, there are therefore at least $|P| + 1$ red edges between $V^*(Q)$ and $V(P)$, so some vertex in $V(P)$ has red edges to at least two vertices in $V^*(Q)$. The $(\ell - 2)$ path between these vertices in Q is also red, because $\phi|_Q = q_1$, and so we have a red copy of $H_\mathrm{r} = C_\ell$. This contradicts the fact that ϕ is a good colouring. □

5.3. **The graph $\mathscr{G}(\Pi)$.** For any family $\Pi = (\pi_1, \ldots, \pi_m)$ of subsets of $[m]$ construct $\mathscr{G}(\Pi)$ from $\mathscr{G}$ as follows. For each $i, j \in [m]$, complete P^i and Q^j to a copy of G_0 if $i \in \pi_j$, and complete P^i and Q^{m+j} to a copy of G_0 otherwise. Let $M^{ij} = M^{ij}(\Pi)$ be the edges between P^i and Q^j or Q^{m+j}, and let $M(\Pi) = \mathscr{G}(\Pi) \setminus \mathscr{G}$ be the union of all the M^{ij}.

Claim 5.5. *$\mathscr{G}(\Pi)$ is Ramsey for $(H_\mathrm{b}, H_\mathrm{r})$.*

Proof. Assume that $\mathscr{G}(\Pi)$ has a good colouring ϕ. By Proposition 4.3 there are some α and β in $[m]$ so that ϕ restricts on P^α to p_1 and on Q^β and $Q^{m+\beta}$ to q_1. As one of Q^β and $Q^{m+\beta}$ have been completed with P^α to a copy of G_0 in $\mathscr{G}(\Pi)$, this copy of G_0, with R, induces a copy of G_1. This contradicts Claim 5.4. □

Claim 5.6. *For any $\alpha, \beta \in [m]$, $\mathscr{G}(\Pi) \setminus M^{\alpha\beta}$ has a good colouring.*

Proof. By taking an automorphism of $\mathscr{G}(\Pi)$, we may assume that $\alpha = \beta = 1$. Let ϕ be the good colouring of $\mathscr{G}$ from Proposition 4.3 (ii) for this choice of α and β. This colouring ϕ is safe on $S_{\mathscr{G}}$, so to show that there is a good colouring of $\mathscr{G}(\Pi)$ it is enough to show that the restriction of ϕ to $S_{\mathscr{G}}$ can be extended to a good colouring of $S_{\mathscr{G}} \cup (M(\Pi) \setminus M^{11})$.

Let v_* be a vertex in P^1 and let ϕ be red on edges in $M(\Pi) \setminus M^{11}$ that are incident to v_*, and blue on those that are not.

The colouring ϕ decomposes $S_{\mathscr{G}} \cup (M(\Pi) \setminus M^{11})$ into the subgraph S_b of edges that are blue under ϕ and the subgraph S_r of edges that are red under ϕ. To show ϕ is good, we show that S_b contains no copy of H_b and S_r contains no copy of H_r.

Observe that S_b is the union of

(i) P^1,
(ii) proper subgraphs of P^i for $i \geq 2$,
(iii) stars centered at the vertices of $V^*(\mathscr{Q})$, (sharing endpoints if $\ell = 4$),
(iv) edges of $M(\mathscr{I})$ not incident to v^*, and
(v) R and $R_{\mathscr{Q}}$.

The set $I = V^*(\mathscr{Q}) \cup \{v_*\}$ is an independent set in S_b. With respect to I the odd components of S_b are $P^1 \setminus \{v^*\}$, proper subgraphs of P^i, and odd components of R with respect to I. Those in the P^i are strictly smaller than P, as they are proper subgraphs of copies of P, so have size less than m_b. In R, there are $M_b - 1$ odd components of size m_b. So by Claim 5.3, S_b cannot contain H_b as a subgraph.

Now, S_r consists of several disjoint graphs:

(i) Q^1 and Q^{m+1},
(ii) disjoint $\ell - 4$ paths in the other copies of Q in $\mathscr{Q}$,
(iii) a star in $M(\mathscr{I})$ with center v_*, and
(iv) one edge in each of the P^i for $i \geq 2$.

The components Q^1 and Q^{m+1} each have girth $3(\ell-2) > \ell$ so cannot contain a copy of H_r. The other components are trees, so cannot contain a copy of H_r. So S_r contains no copy of H_r. This completes the proof that ϕ is a good colouring of $\mathscr{G}(\Pi) \setminus M^{\alpha\beta}$. □

5.4. **Summary of Proof of Theorem 5.1.** For fixed graphs H_b and H_r as in the statement of the proposition, we use Proposition 4.3 in Subsection 5.1 to find a constant c, and for fixed $m \geq 3$ we define the graph $\mathscr{G}$ on at most cm vertices.

For each of 2^{m^2} choices of the family Π of m subsets of $[m]$ we constructed a super graph $\mathscr{G}(\Pi)$ of $\mathscr{G}$ by adding only edges, which by Claim 5.5 is Ramsey for $(H_\mathrm{b}, H_\mathrm{r})$. We claim that any Ramsey-minimal subgraphs of $\mathscr{G}(\Pi)$ and $\mathscr{G}(\Pi')$, for different Π and Π', are different. Indeed, if $\Pi = (\pi_1, \ldots, \pi_m)$ is different from $\Pi' = (\pi'_1, \ldots, \pi'_m)$ then there exist i and j such that, wlog, $i \in \pi_j \setminus \pi'_j$. Thus the sets $M^{ij}(\Pi)$ and $M^{ij}(\Pi')$ are disjoint, and more importantly, $M^{ij}(\Pi)$ is disjoint from $\mathscr{G}(\Pi')$. It follows by Claim 5.6 that any Ramsey-minimal subgraph of $\mathscr{G}(\Pi)$ contains an edge of $M^{ij}(\Pi)$, so is different from any subgraph of $\mathscr{G}(\Pi')$. This proves the claim.

Thus we have at least 2^{m^2} different graphs on at most cm vertices that are Ramsey-minimal for $(H_\mathrm{b}, H_\mathrm{r})$. By Lemma 2.1, this is what we needed, so completes the proof of Theorem 5.1.

6. A Lemma about Hypergraphs

Before we do our final version of the construction from Section 5 we need a standard probabilistic result about hypergraphs. One could also get this constructively using a stronger result about *hypergraph expanders* from [12].

A *d-ary hypergraph* $\mathscr{K}$ is a set of d element subsets, called *hyperedges*, of its vertex set $V(\mathscr{K})$. The *girth* of a hypergraph $\mathscr{K}$ is the minimum g for which there is a sequence

$$v_1, e_1, v_2, e_2, \ldots, v_g, e_g$$

where the e_i are hyperedges and $e_i \cap e_j$ is $\{v_j\}$ if $j = i + 1 (modulo g)$ and is otherwise empty. Such a sequence is called a *g-cycle* in $\mathscr{K}$.

Lemma 6.1. *For a positive constant $c \leq 1$, integers $d, g, > 2$ and a large enough integer n, there is a d-ary hypergraph $\mathscr{K} = \mathscr{K}(c, d, g)$ of girth g, on n vertices, such that any vertex set of size at least cn induces a hyperedge.*

Proof. Let $p = \frac{n^{1/(g-1)}}{n^{d-1}}$ and $\mathscr{H}$ be a random d-ary hypergraph such that each d element subset e of $[n]$ is a hyperedge of $\mathscr{H}$ with probability p, independent of other hyperedges.

The expected number of cycles of length k is

$$\frac{n!}{2k \cdot ((h-2)!)^k (n-(d-1)k)!} \cdot p^k < c_0 \cdot n^{(d-1)k} \cdot p^k = c_0 \cdot n^{k/(g-1)},$$

for constant $c_0 = 2k \cdot ((h-2)!)^k$. So the expected number of cycles of length at most $g-1$ is less than Cn for some constant C independent of n. By the Markov inequality $\mathscr{H}$ has more than Cn cycles of length less than g with probability strictly less than 1.

The expected number of hyperedges induced by a cn element subset V of $[n]$ is

$$\mu = \binom{cn}{d} \cdot p \geq \frac{c^d}{d^d} n^{(1+\frac{1}{g-1})} =: c_1 \cdot n^{(1+\frac{1}{g-1})}.$$

By the Chernoff inequality, the probability that a given set of cn vertices induces less than $Cn = \mu \cdot \frac{Cn}{\mu} = \mu \frac{C}{c_1} \cdot n^{-g/(g-1)} < \mu(1-1/2)$ (for large n) hyperedges, is less than $e^{-\mu/8}$. So the probability that there is some set of cn with at most Cn hyperedges is less than

$$\binom{n}{cn} \exp(-\mu/8) \leq \exp(cn \log n - \mu/8),$$

which is $o(1)$ as $cn \log n$ is $o(\mu)$.

We have then, that with positive probability $\mathscr{H}$ contains at most Cn cycles of length less than g and all sets of cn vertices induce more than Cn hyperedges. Fixing such a graph, we can thus remove all cycles of length less than g by removing one hyperedge from each, without removing all hyperedges induced by any set of cn vertices. This gives the required hypergraph $\mathscr{K}$.

□

7. The Second Case of our Main Construction

In this section, we prove the following.

Theorem 7.1. *Let H_b be non-bipartite, and H_r be a non-complete 3-connected graph such that there is a safe sender for $(H_\mathrm{b}, H_\mathrm{r})$, then $(H_\mathrm{b}, H_\mathrm{r})$ is highly Ramsey-infinite.*

Since the senders constructed for 3-connected graphs in [10] are automatically safe, and pairs of cliques were shown to be highly Ramsey-infinite in [16], we immediately get the following, our main result.

Corollary 7.2. *Let H_b and H_r be 3-connected graphs. Then $(H_\mathrm{b}, H_\mathrm{r})$ is highly Ramsey-infinite if and only if one at least one of them is non-bipartite.* □

We now start the proof of Theorem 7.1. Let H_b and H_r be as in the statement of the proposition, with the further assumption that $(H_\mathrm{b}, H_\mathrm{r})$ is asymmetric. The proof of the asymmetric case closely follows the proof of Theorem 5.1 and so we refer back to that proof for much of it. We keep our numbering consistent with that proof. At the end of the section we observe how the assumption that $(H_\mathrm{b}, H_\mathrm{r})$ is asymmetric can easily be dropped.

7.1. **Definition of $\mathscr{G}(P, Q, R, V^*(Q), p_0, p_1, q_0, q_1, m)$.** Let I_b, m_b and M_b, as well as P, R, p_0, and p_1 be as in Subsection 5.1.

7.1.3. *Definition of Q, q_0, and q_1.* As H_r is not a clique, there is some vertex v with a non-neighbour v'. Let $F = H_\mathrm{r} \setminus v$, and let $N \subset F$ be the neighbourhood of v in H_r. Let $c = 1/(2|P|)$, $d = |N|$, $g = |H_\mathrm{r}| + 1$, and let $\mathscr{K}$ be the hypergraph $\mathscr{K}(c, d, g)$ from Lemma 6.1. For every hyperedge e of $\mathscr{K}$ let F^e be a copy of F and N^e the copy of N in F^e. Construct Q from the set $V^*(Q) = V(\mathscr{K})$ of vertices of $\mathscr{K}$, and the $|\mathscr{K}|$ copies F^e of F by identifying the vertices of N^e with the vertices in e for each e in $\mathscr{K}$:

$$Q = \left((\bigcup_{e \in \mathscr{K}} F^e) \vee_{\mathscr{I}} V(\mathscr{K}) \right)$$

where $\mathscr{I} = \{V(N^e) = e \mid e \in \mathscr{K}\}$. Let q_1 be the edge colouring of Q that colours every edge red. Let q_0 be the colouring of Q that is blue on edges with exactly one endpoint in $V^*(Q)$, and red on every other edge. As F is connected, and $v' \notin N$, q_0 has at least one blue edge in each copy F^e of F in Q. (This is why we need that H_{r} is not complete.)

Remark 7.3. Observe that by the girth $g = |H_{\mathrm{r}}| + 1$ of $\mathscr{K}$, any 2-connected subgraph of Q not fully contained within one of the copies of F, has more than $|H_{\mathrm{r}}|$ vertices.

7.2. **Auxiliary graphs G_0 and G_1.** Given the graphs P and Q and R and the subset $V^*(Q)$ of $V(Q)$, construct G_0 and G_1 exactly as in Subsection 5.2.

Claim 7.4. *There is no good colouring ϕ of G_1 with $\phi|_P = p_1$ and $\phi|_Q = q_1$.*

Proof. The proof is almost the same as that of Claim 5.4. Indeed in exactly the same way we showed there were $|V^*(Q)| - |I_b| + 1 = |P| + 1$ red edges between $|V^*(Q)|$ and $V(P)$ in Claim 5.4, we can show there are $|V^*(Q)| - |I_b| + 1 = |V(\mathscr{K})| - |I_b| + 1$ red edges between $V^*(Q)$ and $V(P)$. Assuming that the number $n_{\mathscr{K}}$ of vertices in $\mathscr{K}$ is large enough, this is at least $n_{\mathscr{K}}/2$ red edges, so some vertex v^* in $V(P)$ has red edges to at least $n_{\mathscr{K}}/(2|P|) = cn_{\mathscr{K}}$ vertices in $V^*(Q) = V(\mathscr{K})$. By the properties of $\mathscr{K} = \mathscr{K}(c,d,g)$ ensured by Lemma 6.1, these vertices induce a hyperedge e of $\mathscr{K}$. As $\phi|_Q = q_1$, ϕ is red on F^e, which with v^* makes up a red copy of H_{r}. This is a contradiction. □

7.3. **The graph $\mathscr{G}(\Pi)$.** Let $\mathscr{G} = \mathscr{G}(P, Q, V^*(Q), R, p_0, p_1, q_0, q_1, m)$ with subgraph $S_{\mathscr{G}}$ be the graph constructed in Proposition 4.3. For $m \geq 3$ and a family $\Pi = (\pi_1, \ldots, \pi_m)$ of subsets of $[m]$, construct $\mathscr{G}(\Pi)$ from $\mathscr{G}$ by adding edges exactly as in Subsection 5.3.

The following follows from Claim 7.4 exactly as Claim 5.5 follows from Claim 5.4.

Claim 7.5. *$\mathscr{G}(\Pi)$ is Ramsey for H_{b} and H_{r}.*

Further, we can show the following.

Claim 7.6. *For any $\alpha, \beta \in [m]$, $\mathscr{G}(\Pi) \setminus M^{\alpha\beta}$ has a good colouring.*

Proof. The proof of this is again similar to the proof of Claim 5.6. In fact, everything is exactly the same up until the point where we have to show that the subgraph S_r of $S_{\mathscr{G}} \cup (M(\Pi) \setminus M^{11})$ of edges that are red under the colouring ϕ, contains no copies of H_{r}. We take the proof from here.

The graph S_r consists of:

(i) Q^1 and Q^{m+1}
(ii) edges in the other Q^i with no endpoints in $V^*(Q^i)$
(iii) edges in the other Q^i with both endpoints in $V^*(Q^i)$,
(iv) edges in $M(\Pi)$ incident to $v_* \in V(P^1)$, and
(v) one edge in each of the copies of P in $\mathscr{P}$ except P^1.

The graphs in (iii) and (iv) make up one big component of S_r. The graphs in (i), (ii) and (v) each consist of smaller components of S_r, so we deal with them first.

The graph (v) consists of independent edges, so cannot contain a copy of H_{r}. As H_{r} is 3-connected and the graph F is a subgraph of H_{r}, it follows by Remark 7.3 that neither Q^1 or Q^{m+1} in (i) contain a copy of H_{r}. The graph in (ii) consists of components each contained in a copy of F. As F is a proper subgraph of H_{r}, they cannot contain a copy of H_{r}.

Now, we have only to show that the union G of the graphs in (iii) and (iv) contains no copy of H_{r}. Assume it contains a copy H of H_{r}. As removing v^* disconnects G into the copies of $V^*(Q)$, H must be contained in the graph induced by $\{v^*\} \cup V^*(Q^i)$ for some

i. But $H \setminus \{v^*\}$ is two connected, and has $|V(H_\mathrm{r})| - 1$ or $|V(H_\mathrm{r})|$ vertices. On the other hand, for any copy F^e of F in Q^i, the copy of v' in F^e is not in $V^*(Q^i)$, so $|V(F^e \cap Q^i)| \leq |V(H_\mathrm{r})| - 2$ vertices. Thus by Remark 7.3, $H \setminus \{v^*\}$ has more than g vertices, contradicting the fact that $|H| = g$. So G contains no copy of H_r.

This shows that S_r contains no copy of H_r, so completes the proof. □

7.4. **The Symmetric Case.** In the case that H_b and H_r are isomorphic, determiners cannot exist, but we can replace them with senders in the following way. Let $A = (A, e_b, e_r)$ be a safe negative sender. We may assume, by permuting colours, that under any good colouring ϕ of A we have $\phi(e_b) = b$ and $\phi(e_r) = r$. In any construction in the paper, in which we attach a safe blue (resp. red) determiner to an edge e we can instead attach e and e_b (resp. e_r) with a safe positive sender. The proofs all hold.

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E-mail address: mhsiggers@knu.ac.kr

COLLEGE OF NATURAL SCIENCES, KYUNGPOOK NATIONAL UNIVERSITY, 1370 SANKYUK-DONG, BUK-GU, DAEGU, 702-701, SOUTH KOREA, TEL: +82-53-950-6309 FAX: +82-53-950-6306